

ARP Configuration

using Cisco Packet Tracer

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Objective



Understand ARP Protocol

Gain foundational knowledge of how Address Resolution Protocol works within IP networks.



Learn IP to MAC Mapping

Understand how IP addresses are mapped to MAC addresses using broadcast and unicast mechanisms.



Implement ARP in Cisco Packet Tracer

Simulate a real LAN environment and trigger ARP resolution between connected devices.



Observe ARP Packets

Use Packet Tracer's Simulation Mode to analyze ARP request and reply packets step-by-step.

Introduction to ARP



ARP

Address Resolution Protocol

RFC 826 — Published 1982
Core Protocol of TCP/IP Suite
Operates at Network Layer



Protocol Layer

Network / Data Link Bridge



Function

Maps IP address → MAC address



Scope

Works within a single LAN segment



Trigger

Automatically called before each packet send

Why ARP is Needed?



Layer 3: IP Address

Computers on the internet identify each other using logical IP addresses (e.g. 192.168.1.2). IP addresses are used for routing packets across networks.

Layer 2: MAC Address

The actual data transmission on a LAN uses physical MAC addresses burned into network interface cards — e.g. AA:BB:CC:DD:EE:FF.

The Gap Problem

IP packets must be delivered using MAC frames at the Data Link Layer. Devices don't automatically know which MAC corresponds to a given IP.

ARP Bridges the Gap

ARP allows a device to broadcast 'Who has IP X?' and receive the MAC address in reply — enabling correct frame delivery on the LAN.

ARP Working Process



ARP Request (Broadcast)

PC-A needs MAC for 192.168.1.3.
Sends broadcast frame to FF:FF:FF:FF:FF:FF
→ All devices on LAN receive it.



Broadcast

ARP Reply (Unicast)

PC-C recognizes its own IP.
Responds directly to PC-A with its MAC address.
→ Only PC-A receives the reply.

Unicast Reply

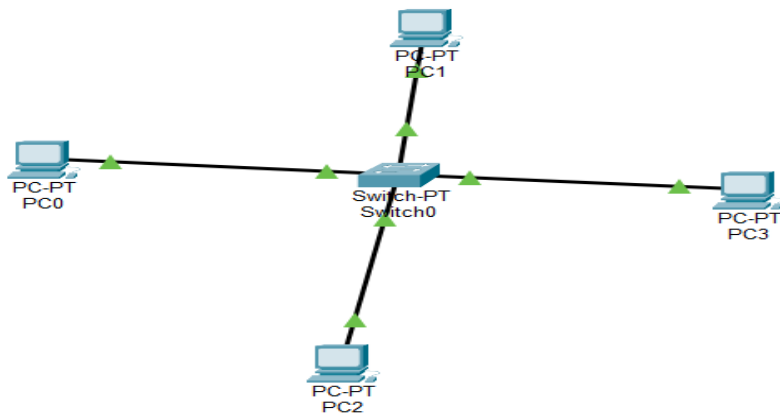
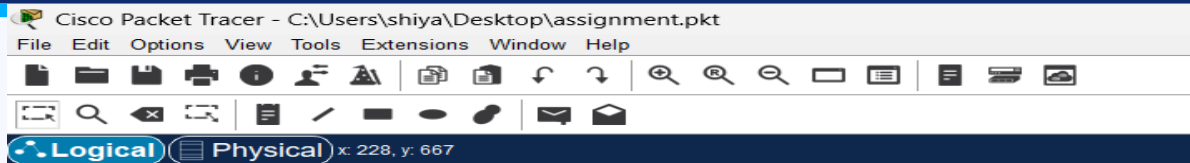
MAC Stored in ARP Cache

PC-A stores the IP→MAC mapping in its ARP table.
All future packets to 192.168.1.3 use cached MAC.
→ No more broadcast needed.



ARP Cache

Network Topology



Network Info

Network	192.168.1.0 / 24
Subnet Mask	255.255.255.0
Devices	4 PCs + 1 Switch
Topology	Star
Medium	Ethernet (Copper)
Simulator	Cisco Packet Tracer

Device Configuration



Device	IP Address	Subnet Mask	Gateway	Port
PC-0	192.168.1.1	255.255.255.0	N/A	Fa0/1
PC-1	192.168.1.2	255.255.255.0	N/A	Fa0/2
PC-3	192.168.1.3	255.255.255.0	N/A	Fa0/3
PC-2	192.168.1.4	255.255.255.0	N/A	Fa0/4

All PCs share the same subnet 192.168.1.0/24 — No default gateway required for intra-LAN communication

Commands Used



The image displays four screenshots of PC configuration windows, each showing the 'Desktop' tab under the 'Config' menu. Each window is for a different PC (PC0, PC1, PC2, PC3) and shows the 'IP Configuration' section for the 'FastEthernet0' interface. The 'Static' radio button is selected for all PCs. The IP addresses and subnet masks are highlighted with blue hand-drawn boxes.

PC	Interface	IP Configuration Type	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	FastEthernet0	Static	192.168.1.1	255.255.255.0	0.0.0.0	0.0.0.0
PC1	FastEthernet0	Static	192.168.1.2	255.255.255.0	0.0.0.0	0.0.0.0
PC2	FastEthernet0	Static	192.168.1.4	255.255.255.0	0.0.0.0	0.0.0.0
PC3	FastEthernet0	Static	192.168.1.3	255.255.255.0	0.0.0.0	0.0.0.0

ARP Table Output



```
PC3
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>arp
Cisco Packet Tracer PC ARP
Display ARP entries: arp -a
Clear ARP table: arp -d

C:\>arp -a
Internet Address    Physical Address    Type
192.168.1.1         000d.bdd1.aeea     dynamic

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a
Internet Address    Physical Address    Type
192.168.1.2         0040.0bea.2137     dynamic
```

Internet Address

192.168.1.2

The target IP address resolved by ARP

Physical Address

0040.0bea.2137

MAC address returned via ARP Reply

Type: Dynamic

Dynamic Entry

Automatically learned — expires after ARP timeout (usually 120–600 sec)

Result & Observation



ARP Successfully Resolved

IP address 192.168.1.3 was correctly mapped to its MAC address using broadcast-based ARP resolution.

Communication Established

After ARP resolution, all 4 PCs were able to communicate using ICMP ping without any packet loss.



No Packet Loss Observed

Ping results showed 100% success — 4 packets sent, 4 received. Round-trip time was minimal within LAN.



ARP Cache Populated

The arp -a command confirmed all neighboring IP-to-MAC mappings stored as dynamic entries in ARP cache.

Advantages of ARP



01 Fast LAN Communication

ARP enables seamless and automatic MAC resolution, allowing near-instant communication between devices on the same LAN segment without manual intervention.

02 Automatic Address Resolution

Devices don't need pre-configured MAC tables. ARP dynamically discovers and caches MAC addresses on demand, simplifying network administration.

03 Reduces Manual Configuration

No need for static ARP entries in most environments. Reduces configuration overhead for network admins and supports plug-and-play device connectivity.

04 Works Transparently

ARP operates silently in the background. End users and applications don't need to be ARP-aware — it integrates fully with the OS networking stack.

Limitations of ARP



LAN-Only Scope

ARP works only within a single broadcast domain. Across routers or VLANs, ARP messages do not travel — routers use routing tables instead.



Vulnerable to ARP Spoofing

Malicious devices can send fake ARP replies, poisoning the ARP cache of other hosts. This enables Man-in-the-Middle (MitM) attacks on the LAN.



No Authentication Mechanism

ARP has no built-in way to verify the source of replies. Any host can claim any IP's MAC address — there is no trust or certificate system.

Conclusion



"ARP is the invisible handshake that makes every LAN conversation possible."



ARP is Essential for LAN

Without ARP, IP-level communication on Ethernet LANs would be impossible. It bridges the gap between Layer 3 and Layer 2 addressing.



Successfully Simulated in Packet Tracer

The complete ARP workflow — request, reply, and cache — was observed and verified using Cisco Packet Tracer simulation mode.



Real-Time Packet Flow Learned

By analyzing event logs and PDU details, we gained hands-on understanding of how ARP packets propagate across a switch-based LAN topology.



Thank You!

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